



Enhancement of thermal and mechanical properties of microencapsulated phase change materials with graphene oxide

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ABSTRACT

The low thermal conductivity of microencapsulated phase change materials (MEPCMs) limits the latent heat charging and discharging rates for various applications. To overcome this limitation, we prepared MEPCM by co-surfactants of polyvinyl alcohol and high thermal conductivity graphene oxide (GO) through an emulsion polymerization process. Fourier transformation infrared spectroscopy and Raman spectra results verified that GO was successfully added to the MEPCMs' hybrid polymer shell. The core material content of MEPCM and GO/MEPCM was within the range of 78.4 %–91.8 % according to differential scanning calorimetry testing results. According to the comparison of fabricated microcapsule samples, the dosage of 0–0.5 w.t.% GO can reduce the loss of shell monomers and overcome the supercooling and leakage problem of MEPCM. Thermogravimetric results exhibited that the thermal stability of MEPCM samples increased by 66 °C after encapsulation, and this value further increased by 7–21 °C with the addition of 0.1–0.5 w.t.% GO. The thermal conductivity of MEPCM samples increased from 0.32 W/m•K to 1.04 W/m•K with a dosage of 0.5 w.t.% GO. Meanwhile, Young's modulus and the hardness of GO/MEPCM samples with 0.5 w.t.% GO increased by 0.2 GPa and 0.1 GPa, respectively.

1. Introduction

Microencapsulated phase change materials (MEPCMs) can effectively solve the leakage problem, enlarge the heat transfer area, and reduce the influence of the external environment and the volume change of traditional solid–liquid phase change materials (PCMs) [1,2]. Therefore, MEPCMs can be widely applied in various applications, such as solar energy, waste heat recovery [3], energy conservation in buildings [4,5], reduction of CO₂ emission [6], textile products [7] etc. However, the low thermal conductivity of MEPCMs (usually between 0.15 and 0.4 W/m • K [8]) limits the heat storage/release rates from these materials in most applications. At the same time, improving the

mechanical strength of MEPCMs is also important for embedding mixtures of MEPCM particles into composite materials [9].

Previous studies have shown that nanomaterials such as graphene, carbon nanotube and Fe₃O₄ [10] can enhance the thermal performance of MEPCM. For example, Xiong et al. [11] achieved 44.5–50.1 % increase in the thermal conductivity of cement paste incorporating polystyrene/n-eicosane MEPCM with 5 w.t.% of graphite platelets. Khezri et al. [12] fabricated MEPCM with 10 w.t.% of graphene nanoplatelets and showed that graphene was able to enhance the thermal conductivity by about 48 % (0.29 W/m•K vs. 0.43 W/m•K). Chen et al. [13] used graphene nanosheets and regenerated nano chitin as a Pickering emulsifier for the development of composite MEPCM. The results demonstrated the thermal conductivity of the MEPCM was enhanced by

Abbreviations: AIBN, Azodiisobutyronitrile; DSC, Differential scanning calorimetry; FTIR, Fourier transform infrared spectroscopy; GO, Graphene oxide; MEPCM, Microencapsulated phase change material; MMA, Methyl methacrylate; PCM, Phase change material; PDMS, Polydimethylsiloxane; PETRA, Pentaerythritol tetraacrylate; PMMA, Poly(methyl methacrylate); PVA, Polyvinyl alcohol; SEM, Scanning electron microscopy; TGA, Thermogravimetric analyzer.

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Nomenclature			
Symbols		ν	Poisson's ratio
C	Core material content (%)	φ	Volume fraction (%)
E	Encapsulation efficiency (%)	Subscript	
H	Hardness (GPa)	a	Air
Lr	Leakage rate (%)	c	Crystallization
P	Load (mN)	ci	Onset of the crystallization process
T	Temperature (°C)	ce	Endset of the crystallization process
Y	Young's modulus (GPa)	e	Effective
k	Thermal conductivity (W/m • K)	m	Melting
ΔH	Latent heat enthalpy (J/g)	mi	Onset of the melting process
ΔT	Supercooling degree (°C)	me	Endset of the melting process
ρ	Density (kg/m ³)	$m100$	100 heating-cooling cycles
		r	Real

140 % with 10 w.t.% graphene (0.27 W/m•K before vs. 0.65 W/m•K after the addition of graphene). In a study where both mechanical and thermal properties of MEPCM samples with graphene were investigated, Su et al. [14] compared the mechanical strength of fabricated MEPCM samples with graphene/organic hybrid shell and organic shell without graphene. The measurements demonstrated that Young's modulus and yield stress limit of MEPCM were increased by about 94 % and 42 % respectively for the sample that had 6 w.t.% of graphene according to nanoindentation testing. Meanwhile, the thermal conductivity of MEPCM was increased by 100 % for the sample with 2 w.t.% graphene.

On the other hand, amphiphilic graphene oxide (GO), which contains hydroxyl, epoxy groups and carboxyl groups, can be used as a surfactant for stabilizing oil and water emulsions [15]. This was demonstrated by Zhang et al. [16] who combined calcium lignosulfonate with aqueous GO to form an emulsifier. In this way, a novel linseed oil self-healing microcapsule was successfully developed within a GO/polyaniline hybrid shell by the Pickering emulsion template method. In addition, Zhang et al. [17] prepared and enhanced the leakage-prevention performance of MEPCM by emulsifying paraffin in water with amphiphilic GO as a surfactant. The experimental results demonstrated that the amphiphilic GO was successfully placed at the interface between the PCM core and poly (melamine-formaldehyde) shell, thus decreasing the leakage rate of the MEPCM by 93.1 % compared with the sample without GO. Jiang et al. [18] fabricated MEPCM with GO and CaCO₃ as hybrid shells and paraffin as core material to evaluate their mechanical properties. In this study by Jiang et al. [18], the CaCO₃ shell was fabricated by the reaction of CaCl₂ solution and Na₂CO₃ solution. The testing results revealed a reduction of 54 % in the breakage rate of the microcapsule by using centrifugal acceleration testing when 1.0 w.t. % of GO was added to it. In addition, the thermal conductivity of the MEPCM samples was also improved by adding 1.5 w.t.% GO which increased from 0.72 W/m • K to 0.88 W/m • K. The above studies showed the significant advantages of microcapsules fabricated with GO, such as higher thermal conductivity, and better performance in terms of mechanical strength, thermal stability, tightness and impermeability. Thermal and mechanical properties improvement of MEPCM would also contribute to enhancements of the performance of its composite materials, i.e. MEPCM could be integrated with polymeric matrices and be used for temperature management and thermal energy storage [19], textiles for personal heat management [20] and concrete composites in buildings for energy saving [21] etc.

Polymethyl methacrylate (PMMA) based polymers were often used as shell materials for the fabrication of MEPCMs [22,23]. This was because PMMA has the advantages of high sphericity, thermal stability, mechanical strength, non-toxicity and can be flexibly initiated by many polymerization methods for MEPCM preparation [24,25]. Meanwhile, the introduction of crosslinking agents for the polymerization of methyl methacrylate (MMA), such as divinylbenzene [26], allyl methacrylate

[27], methacrylic acid [28], and pentaerythritol tetra-acrylate (PETRA) [29] can improve the quality of MEPCM. For example, Al-Shannaq et al. [30] studied the morphology of microcapsules after adding crosslinking agent PETRA and found that the surface morphology of MEPCM was smoother and denser.

In order to obtain high thermal conductivity MEPCMs with better thermophysical and mechanical properties, this study proposed to use GO and polyvinyl alcohol (PVA) as composite emulsifiers for the preparation of the MEPCM capsules. MMA was used as the shell monomer and PETRA crosslinking agent for the fabrication of GO/PMMA-PETRA hybrid shells to encapsulate the PCM core by emulsion polymerization. Moreover, the effect of the dosage of GO on the morphology, thermal conductivity, phase transition characteristics and thermal stability of prepared MEPCM samples was also evaluated.

2. Materials and methods

2.1. Raw materials

In this study, n-octadecane (purity 99 %, Shanghai Macklin Biochemical Co., Ltd.) was used as a PCM. MMA (purity 98 %, Sino-pharm Chemical Reagent Co., Ltd.) and PETRA (Shanghai Macklin Biochemical Co., Ltd.) were used as the shell monomer and crosslinking agent, respectively. PVA-203 (Molecular weight ~ 31000, Shanghai Aladdin Biochemical Technology Co., Ltd.) and GO suspension (5 mg/mL, Jining Leadernano Tech. L.L.C.) were used as emulsifiers. The dosage percentage of GO during the MEPCM samples preparation was based on the initial theoretical weight of MEPCM samples. For example, GO/MEPCM-1 contained 0.1 w.t.% of GO which used 2 mL of 5 mg/mL GO suspension. The thermal initiator azodiisobutyronitrile (AIBN) (Shanghai Macklin Biochemical Co., Ltd.) was used to initiate the polymerization of MMA and PETRA monomers. Hydrochloric acid was used to adjust the pH of the GO oil-water emulsion. The detailed technical specifications of raw materials for various MEPCM fabrications are summarized in Table 1.

2.2. Preparation method of MEPCM samples

The fabrication process of MEPCM is shown in Fig. 1. Firstly, MMA monomer, PETRA crosslinking agent and molted n-octadecane were mixed with thermal initiator AIBN at 40 °C to form the oil phase. Then, 10 mL of PVA solution with a concentration of 0.05 g/mL was taken in a beaker and diluted with 60 mL of deionized water to obtain the aqueous phase. Different weight fractions of GO suspension, as listed in Table 1, were added into the PVA solution by ultrasonic dispersion (Ningbo Scientz Biotechnology Co., Ltd) for 10 min as the water phase. The pH of the water phase solution was adjusted to 6 with hydrochloric acid. The mixed oil phase and water phase were emulsified with a highspeed

Table 1

Raw materials for MEPCM fabrications.

Sample	MMA (g)	PETRA (g)	PCM (g)	AIBN (g)	PVA (g/mL)	GO (w. t.%)
MEPCM	2.7	0.3	7.0	0.06	0.005	0
GO/MEPCM-1	2.7	0.3	7.0	0.06	0.005	0.1
GO/MEPCM-2	2.7	0.3	7.0	0.06	0.005	0.2
GO/MEPCM-3	2.7	0.3	7.0	0.06	0.005	0.3
GO/MEPCM-4	2.7	0.3	7.0	0.06	0.005	0.4
GO/MEPCM-5	2.7	0.3	7.0	0.06	0.005	0.5

disperser (IKA® - Werke GmbH & Co. KG) at a stirring rate of 5000 rpm for 5 min to form a stable oil-in-water emulsion which was then transferred into a flask and deoxygenated with nitrogen for 10 min. The mixture's temperature was adjusted to 70 °C for 2 h at a stirring rate of 500 rpm and then raised to 80 °C at the same rate for another 3 h for the microencapsulation reaction. Finally, the MEPCM samples were washed 3 times with water at 70 °C, filtered and then dried over 24 h at 60 °C.

2.3. Characterization of the MEPCM samples

Fourier transform infrared spectroscopy (FTIR) (BRUKER ALPHA, Germany) was used to analyze the chemical composition of GO, PMMA-PETRA, n-octadecane, MEPCM and GO/MEPCM. The tests were carried out at room temperature within a test range of 4000–500 cm⁻¹. Raman spectra (HR800, HORIBA JOBIN YVON, France) was used to confirm GO into GO/MEPCM samples with an excitation wavelength of 514.5 nm at a testing range of 1000–1800 cm⁻¹.

The microstructure of the microcapsules was obtained with scanning electron microscopy (SEM) (Phenom Prox, PHENOM SCIENTIFIC). All MEPCM samples were coated with a 2 nm gold layer before observation.

Differential scanning calorimetry (DSC) (DSCQ20, TA Instruments) was used to test the latent enthalpies and phase change temperatures under the nitrogen-protected environment with a cooling/heating rate of 2 °C/min from 0 °C to 50 °C. According to DSC measurement results, the core material content (C) and encapsulation efficiency (E) of the MEPCM samples were calculated by Eq. (1) and Eq. (2). Meanwhile, Eq. (3) and Eq. (4) were used to calculate the theoretical shell thickness of the MEPCM based on the actual shell-core ratio obtained by Eq. (1).

$$C = \frac{[\Delta H_{m(MEPCM)} + \Delta H_{c(MEPCM)}]/2}{[\Delta H_{m(PCM)} + \Delta H_{c(PCM)}]/2} \quad (1)$$

$$E = \frac{[\Delta H_{m(MEPCM)} + \Delta H_{c(MEPCM)}]/2}{[\Delta H_{m(PCM)} + \Delta H_{c(PCM)}]/2 \times (W_{core}/(W_{core} + W_{shell}))} \quad (2)$$

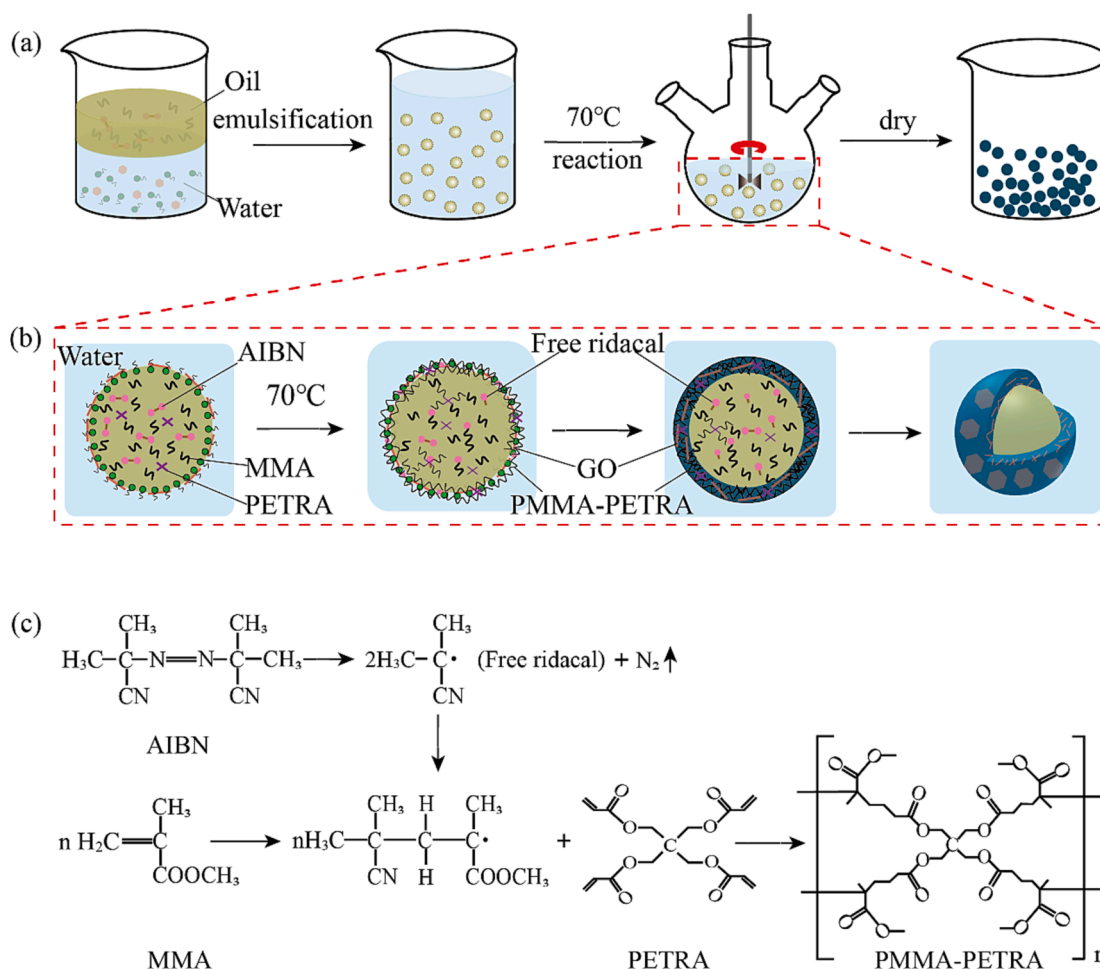


Fig. 1. Process diagram of MEPCM preparation. (a) Preparation process of MEPCM. (b) Shell formation mechanism of MEPCM. (c) Free radical polymerization.

$$\frac{r^3}{R^3 - r^3} = \frac{C}{1 - C} \quad (3)$$

$$d = R - r \quad (4)$$

where $\Delta H_{m(\text{MEPCM})}/\Delta H_{c(\text{MEPCM})}$ and $\Delta H_{m(\text{PCM})}/\Delta H_{c(\text{PCM})}$ represent the melting/crystallization latent heat of MEPCM and PCM core (n-octadecane), respectively. W_{core} and W_{shell} represent the corresponding mass fractions of the core and shell. R/r is the radius of MEPCM/core material, and d represents the theoretical shell thickness.

The thermal stabilities were evaluated with a thermogravimetric analyzer (TGA) (TGA2, Mettler Toledo) by measuring the samples under the protection of nitrogen (N_2) within a heating range of 40–500 °C and a heating rate of 10 °C/min.

The anti-leakage performance of the MEPCM samples was tested by 100 solidification and melting cycles with a self-made semiconductive thermoelectric panel for cooling (to 10 °C) and heating (to 50 °C). The leakage rate (Lr) of the MEPCM sample was then calculated by the reduced melting enthalpy after 100 thermal cycles against its initial latent heat, as shown in Eq. (5).

$$Lr(\%) = \frac{\Delta H_m - \Delta H_{m100}}{\Delta H_m} \times 100\% \quad (5)$$

The thermal conductivity of PMMA-PETRA and MEPCM samples were measured with a thermal conductivity meter (DZDR-S, Nanjing Dazhan Mechanical and Electrical Technology Research Institute) through transient plane heat source method at the same temperature to ensure that the PCM was at a consistent phase. However, the measured values were affected by the porosity of the samples. Therefore, the real thermal conductivity of the samples was calculated according to the parallel model of Kohout et al. [31], as shown in Eq. (6) and Eq. (7):

$$k_e = \varphi_r \bullet k_r + \varepsilon \bullet k_a \quad (6)$$

$$\varphi_r + \varepsilon = 1 \quad (7)$$

$$\varphi_{\text{shell}} = \frac{(1 - C)/\rho_{\text{shell}}}{\frac{1 - C}{\rho_{\text{shell}}} + \frac{C}{\rho_{\text{pcm}}}} \quad (8)$$

k_e , k_r and k_a represent the effective thermal conductivity, real thermal conductivity and air thermal conductivity (0.0267 W/mK), respectively. φ_r and φ_{shell} represent the volume fraction of the MEPCM and its shell, ε represents the porosity, and ρ_{shell} and ρ_{pcm} represent the density of the MEPCM and the PCM core. The porosity ε was tested and calculated through Archimedes' drainage method.

The mechanical properties of MEPCMs were tested by nano-indentation (Keysight G200, Germany), following the same method as Huang et al. [32] and Guo et al. [33]. In order to fix the MEPCM particles to the testing substrate, polydimethylsiloxane (PDMS) was spin-coated on the surface of the acrylic plate at a speed of 9000 rpm by a spin-coating spinner (12A, Zhangqiu crown brand electronic equipment factory) to form a 5 μm PDMS film. The PDMS was heated at 70 °C for 10 min and the dry MEPCM powder was then spread on the surface of the PDMS glue. Finally, continuous heating for 1 h enabled the stable positioning of MEPCM particles on the surface of the PDMS film. The nanoindentation testing was carried out in XP mode and the selected MEPCM particle spherical centre point was used for the measurement under the microscope. The indentation rate and maximum indentation depth of the indenter were set at 10 nm/s and 5000 nm with a holding time of 10 s. Hardness (H) and Young's modulus (Y) were calculated from formulas 9, 10 and 11 [32]:

$$H = \frac{P_{\text{max}}}{A} \quad (9)$$

$$Y_r = \frac{1 - \nu^2}{Y} - \frac{1 - \nu_0^2}{Y_0} \quad (10)$$

$$Y_r = \frac{\sqrt{\pi}}{2} \frac{dP}{dH} \frac{1}{\sqrt{A_r}} \quad (11)$$

where P_{max} is the maximum load, A is the projected contact area, Y_r is the reduced modulus, ν is the measured Poisson's ratio of the material, and Y_0 and ν_0 are Young's modulus and Poisson's ratio of the indenter material, respectively.

3. Results and discussion

3.1. Material composition

The FTIR test results for GO, n-octadecane, PMMA-PETRA, MEPCM and GO/MEPCM are shown in Fig. 2 (a), and the absorption peaks are summarized in Table 2. For the raw materials of PMMA-PETRA shell, n-octadecane core and GO, some particular absorption peaks can be used as an identification of themselves. For instance, the strong in-plane rocking vibration absorption peak at 719 cm^{-1} indicates long-chain alkanes with multiple- CH_2 in n-octadecane. The strong absorption peak at 1145 cm^{-1} and 1727 cm^{-1} corresponds to the stretching vibration of C = O and -O- CH_3 in PMMA-PETRA, respectively. The absorption peaks of 3420 cm^{-1} and 1091 cm^{-1} are attributed to the bound O-H and C-OH groups in GO. However, both the PCM core and the polymer shell include a C-H bond which resulted in the stretching vibration absorption peaks of 2912 cm^{-1} and 2848 cm^{-1} and the bending vibration absorption of 1469 cm^{-1} .

In the MEPCM infrared curve, the absorption peaks covered all peaks of the PCM core and PMMA-PETRA shell, as summarized in Table 2. These measurements show that the materials of PMMA-PETRA and n-octadecane were included in MEPCM samples. In the infrared spectrum of GO/MEPCM, in addition to the absorption peak corresponding to the MEPCM, the measurements show an absorption peak of the O-H bond in the combined water of GO at 3420 cm^{-1} . FTIR results showed the characteristic absorption peaks of GO/MEPCM and proved that n-octadecane was successfully encapsulated by a PMMA-PETRA polymer shell and GO was successfully added to GO/MEPCM.

Raman spectroscopy further analyzed GO, MEPCM and GO/MEPCM in Fig. 2 (b). Generally, GO exists at the broad D-band at 1350 cm^{-1} and G-band at 1580 cm^{-1} [34]. Except for MEPCM in Fig. 2 (b), the D-peak and G-peak can be observed in the Raman spectra of GO/MEPCM and GO, which demonstrates the incorporation of GO within the GO/MEPCM microcapsules.

3.2. Morphology of MEPCM samples

The SEM images of the MEPCM and GO/MEPCM samples are shown in Fig. 3. (a) and (c) represent the morphology of MEPCM without GO, and (b), (d) ~ (f) represent the microscopic images of GO/MEPCM samples containing different amounts (0.1 ~ 0.5 w.t.%) of GO. The surface of the sample in Fig. 3 (a) and (c) appears smooth but has some agglomeration problems. This may be due to the shell of the MEPCM being relatively weak and thin. However, the agglomeration issue was significantly reduced in the GO/MEPCM samples which indicates that GO could be used as a surfactant to improve the fabrication rate of MEPCMs. On the other hand, when GO was used as a co-emulsifier with PVA-203, more PMMA-PETRA molecules were accumulating on the surface of the MEPCM. However, the addition of GO increased the surface roughness of the MEPCM, especially for the 0.2 w.t.% GO/MEPCM. Moreover, there are depressions on the surface of MEPCM samples which might be caused by: 1) the density of the polymer shell being higher than the density of initial shell monomers, which results in the volume shrinkage during the polymerization of PMMA-PETRA; 2) the shrinkage stress generated after the solidification of paraffin wax, which changes the pressure balance between the inside and outside of the microcapsules [35].

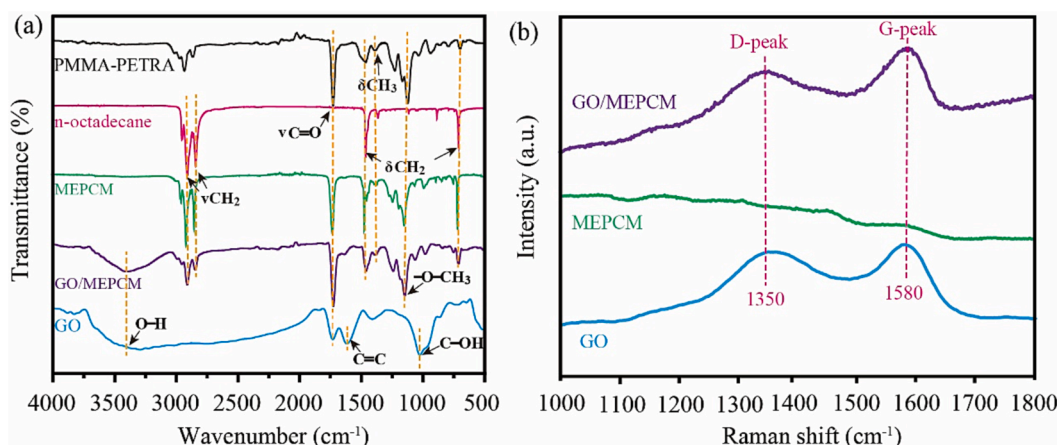


Fig. 2. Composition analysis of experimental samples. (a) FTIR and (b) Raman spectra.

Table 2

FTIR absorption peaks of various materials.

Material	Bond's absorption peaks (cm ⁻¹)					
	O-H	C-OH	C-H	-CH ₂	C=O	-O-CH ₃
n-octadecane	—	—	1469/2912/2848	719	—	—
PMMA-PETRA	—	—	1469/2912/2848	—	1145	1727
GO	3420	1091	—	—	—	—
MEPCM	—	—	1469/2912/2848	719	1145	1727
GO/MEPCM	3420	1091	1469/2912/2848	719	1145	1727

3.3. Phase change temperature and latent heat

The DSC testing results of MEPCM samples prepared with different GO content conditions are shown in Fig. 4 and Table 3. The latent heat range of fabricated MEPCM samples was as high as 191.1–223.3 J/g with 78.4 %–91.8 % core material content. During the melting process, the phase change temperature range of n-octadecane was only 2.58 °C, but this range reached 5.53 °C after encapsulation. This was because the porosity in the MEPCM sample increases the thermal resistance between the PCM core and the external heating/cooling sources due to the existence of air pores between MEPCM particles. Compared with the initial MEPCM sample, the phase change temperature range of GO/MEPCM samples gradually decreased with the increase of GO content. This was because the increase of GO content improves the heat transfer coefficient and thermal conductivity of GO/MEPCM. During the crystallization process, the phase transition temperature range of MEPCM was wider than that of melting. For the MEPCM, due to the limitation of the enclosed space, it was first transformed from an isotropic liquid into a new metastable rotating phase, rather than directly entering the triclinic crystal phase of n-octadecane, which resulted in multiple peaks in the MEPCM crystallization process [36,37]. At the same time, the phase transition temperature range of the MEPCM crystallization process was significantly narrower (by 3–5 °C) than that of the GO/MEPCM. The GO/MEPCM can be more conducive to the polymerization of the monomer during the emulsification process and produce a thicker GO/MEPCM shell, which results in slower crystallization.

Generally, the supercooling temperature (ΔT) was the difference between the onset melting temperature (T_{mi}) and the onset crystallization temperature (T_{ci}). The endset melting temperature and crystallization temperature were recorded as T_{me} and T_{ce} , respectively. According to Table 3, the supercooling temperature of n-octadecane was only 0.79 °C, while the supercooling degree of the MEPCM was

increased to 1.3 °C. The supercooling degree of GO/MEPCM decreased, especially for the sample of 0.3 w.t.% GO/MEPCM where it decreased by 0.63 °C to 0.67 °C.

According to the initial shell-core ratio in Table 1, the theoretical content of the PCM core should be 70 %. However, the DSC test results found that the actual core content of MEPCM was higher than the theoretical value that was derived from Eq. (1). For instance, the actual core content of MEPCM was 21.8 % higher than the theoretical value and shell monomers utilization rate was less than 30 %. This indicated the encapsulation efficiency (E) of MEPCM and GO/MEPCM samples was more than 100 % through the calculation of Eq. (2). Higher core material content and encapsulation efficiency indicate that the shell material content has been lost which would result in some deformation and adhesion problems. The loss of the shell material may be due to two reasons: 1) part of MMA was evaporated during encapsulation due to relatively high reaction temperature; 2) the self-polymerization of MMA in the polymerization process forms nanoparticles and precipitates from the oil phase to the water phase. As the GO content increases, the shell loss of GO/MEPCM is reduced and the core material content of GO/MEPCM is closer to the initial core material ratio in Table 3. It was shown that when GO was added to the MEPCM as an emulsifier, it effectively prevented the diffusion of PMMA-PETRA to the outside of the oil phase during the polymerization process, and improved the utilization rate of the shell material.

3.4. Thermal stability of MEPCMs

The TG and DTG curves of the samples are used to evaluate the thermal stability of MEPCM samples, as shown in Fig. 5. The weight loss of the PCM core of n-octadecane and polymer shell of PMMA-PETRA was just a 1-stage process. For the PCM core in Fig. 5 (a) and (b), it started at about 132 °C and completely disappeared at around 227 °C, mainly due to the volatilization of n-octadecane. At the same time, the initial weight loss temperature of PMMA-PETRA started at 301 °C and was completed at 410 °C as shown in Fig. 5 (a) and (d), due to decomposition of the polymer shell. On the other hand, the weight loss processes of the MEPCM and GO/MEPCM samples followed two weight loss stages of the core material (n-octadecane) and the shell material respectively as Fig. 5 (a) illustrates. Meanwhile, the corresponding DTG curve in Fig. 5 (c) presents the characteristics of a W-shaped curve, which confirms that the weight loss process of the MEPCM sample can be divided into two stages. It can also be seen that the first PCM core evaporation stage is much faster than the second shell decomposition stage.

The initial weight loss temperature of MEPCM was 198 °C, which was 66 °C higher than that of the non-encapsulated n-octadecane. The initial weight loss temperature of GO/MEPCM with 0.1–0.5 w.t.% GO was 210 °C, 219 °C, 208 °C, 205 °C, and 206 °C, respectively. The

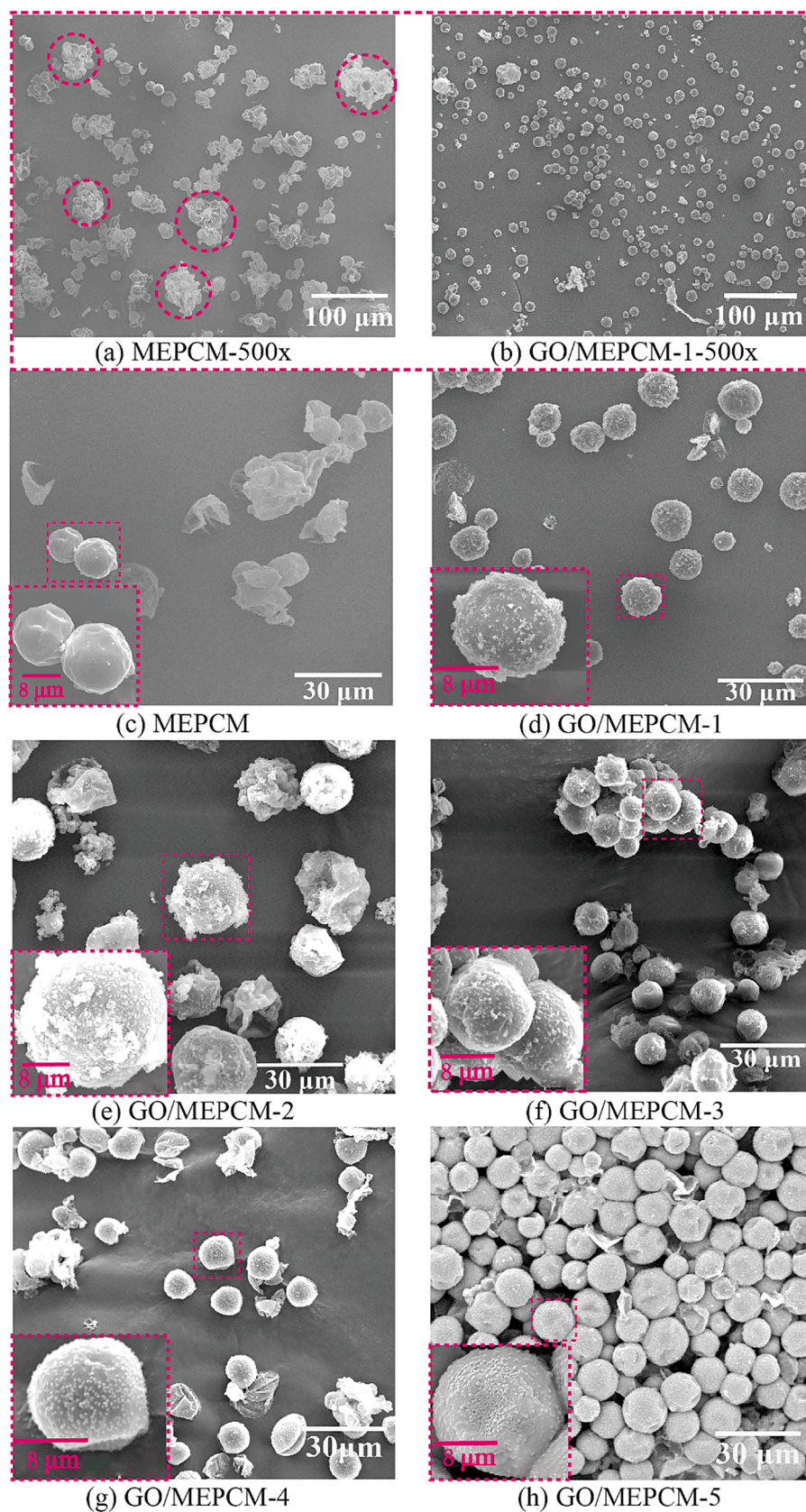


Fig. 3. Morphology images of MEPCM samples by SEM. (a) MEPCM-500x, (b) GO/MEPCM-1-500x, (c) MEPCM, (d) GO/MEPCM-1, (e) GO/MEPCM-2, (f) GO/MEPCM-3, (g) GO/MEPCM-4, (h) GO/MEPCM-5.

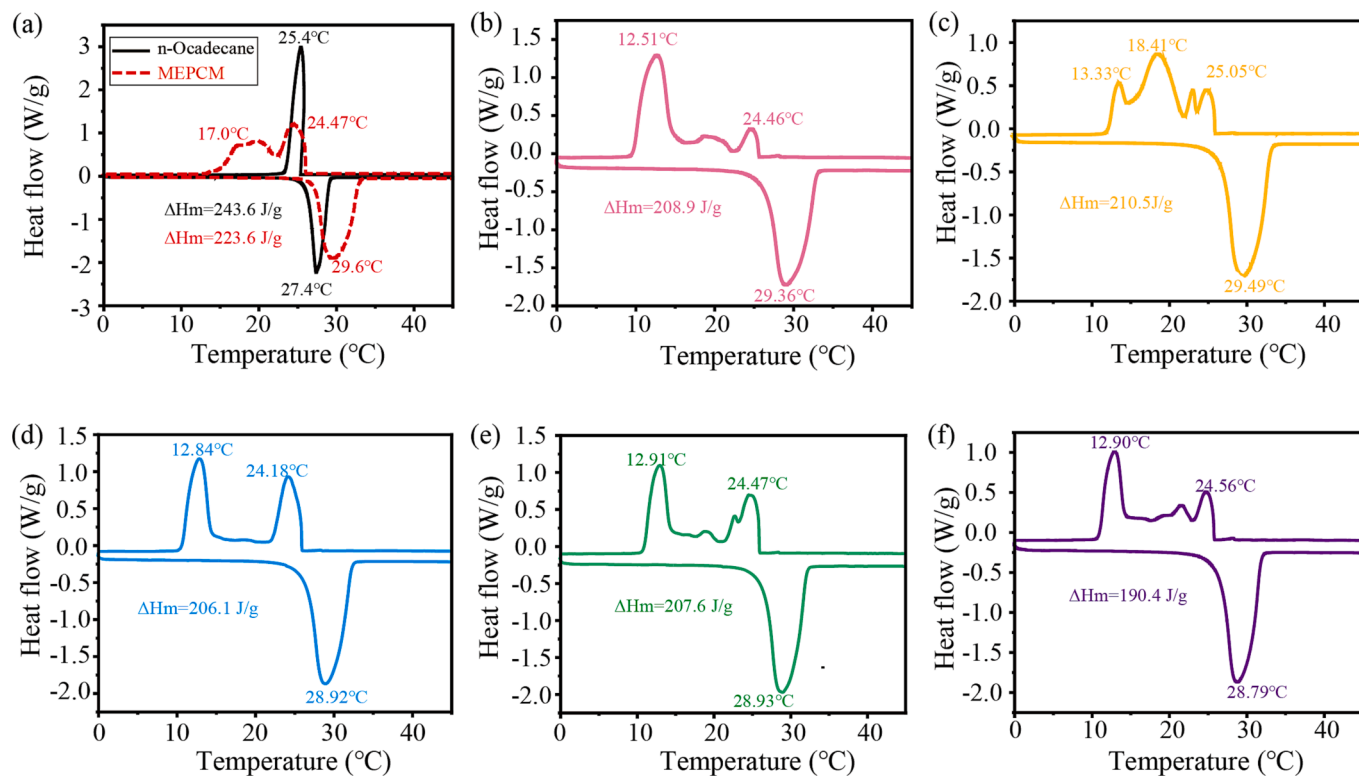


Fig. 4. DSC curves of PCM and MEPCM samples. (a) n-Octadecane & MEPCM. (b) GO/MEPCM-1. (c) GO/MEPCM-2. (d) GO/MEPCM-3. (e) GO/MEPCM-4. (f) GO/MEPCM-5.

Table 3
Summary of thermal properties of MEPCM samples.

Sample	$T_m(^{\circ}\text{C})$		ΔT_m	$\Delta H_m(\text{J/g})$	$T_c(^{\circ}\text{C})$		ΔT_c	$\Delta H_c(\text{J/g})$	$\Delta T(^{\circ}\text{C})$	C (%)	E (%)
	T_{mi}	T_{me}			T_{ci}	T_{ce}					
n-Octadecane	26.15	28.73	2.58	243.6	25.39	23.94	1.45	243.3	0.79	Na.	Na.
MEPCM	27.33	32.86	5.53	223.6	26.03	14.91	11.12	223.3	1.3	91.8	131.1
GO/MEPCM-1	26.64	32.95	6.31	208.9	25.61	9.63	15.98	206.6	1.03	85.3	121.9
GO/MEPCM-2	26.97	33.11	6.14	210.5	25.05	11.05	14.00	210.0	1.92	86.4	123.4
GO/MEPCM-3	26.59	32.25	5.66	206.1	25.92	10.32	15.60	205.5	0.67	84.5	120.8
GO/MEPCM-4	26.66	32.08	5.42	207.6	25.92	10.86	15.06	207.5	0.74	85.3	121.8
GO/MEPCM-5	26.65	31.96	5.31	190.4	25.79	10.90	14.89	191.1	0.86	78.4	111.9

addition of GO further increased the initial weight loss temperature of MEPCM by more than 7 °C. Especially for GO/MEPCM-2, the initial weight loss temperature increased by 21 °C. As the GO content increases in Fig. 5, the slope of the GO/MEPCM also increases probably due to the addition of GO that improves the heat transfer of the GO/MEPCM sample.

3.5. Anti-leakage performance

Anti-leakage testing results are shown in Fig. 6 (a). Six MEPCM samples (about 0.1 g for each powder sample) were put on blotting paper for 100 heating and cooling cycles. Fig. 6 (b)–(d) shows the temperature distribution of MEPCM samples using a thermometer (UT320D, UNI-TREND TECHNOLOGY (CHINA) CO., LTD). The final cooling/heating temperature of 10/50 °C ensured the MEPCM samples were fully solidified/melted. After 100 thermal cycles, the MEPCM samples were collected for DSC testing and the grease on blotting paper was examined, as shown in Fig. 6 (e). It can be seen that some MEPCM samples have oil stains on the underlay paper. Meanwhile, the oil wetting traces of GO/MEPCM samples are significantly decreased by the increase of GO content. In particular, for GO/MEPCM-4 and GO/MEPCM-5, there were almost no observed oil stains on the paper.

The DSC testing results in Fig. 6 (f) after 100 thermal cycles also showed that the anti-leakage performance of GO/MEPCM samples was significantly better than the initial MEPCM. The latent heat for MEPCM, GO/MEPCM-1, GO/MEPCM-2, GO/MEPCM-3, GO/MEPCM-4 and GO/MEPCM-5 was at 200.8 J/g, 196.3 J/g, 199.2 J/g, 201.7 J/g, 203.8 J/g, 189.8 J/g, respectively. According to Eq. (5), the leakage rate of various MEPCM samples was calculated and shown in Fig. 6 (g), and the initial leakage rate of MEPCM was about 10 %. However, the leakage rate of GO/MEPCM gradually decreased with the addition of GO. When the dosage of GO content was at 0.5 wt.%, the leakage rate of GO/MEPCM was only 0.32 %, which confirms that the addition of GO can effectively reduce the leakage of microcapsules. This is because GO as a surfactant makes part of GO form a shell on the surface of the microcapsules, which prolongs the leakage path of the core material and reduces the leakage rate of the microcapsules [17]. On the other hand, a portion of GO was hybridized with the PMMA-PETRA shell, which reduced the shell pores and further increased the anti-leakage performance of the microcapsules.

3.6. Thermal conductivity analysis

Increasing the thermal conductivity is crucial for MEPCMs to achieve

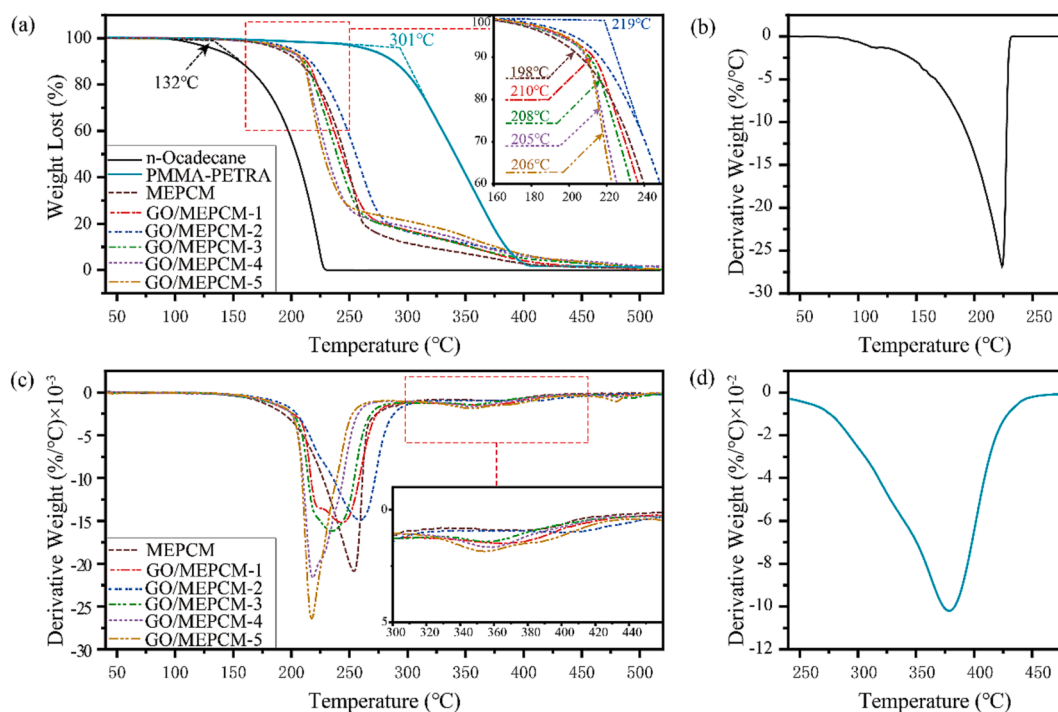


Fig. 5. TG and DTG curves of PCM/MEPCM/shell samples. (a) TG curves of PCM and MEPCM samples. (b) DTG curve of n-Octadecane. (c) DTG curves of MEPCM samples. (d) DTG curve of PMMA-PETRA.

higher thermal energy storage and release capacity. The measured thermal conductivities of n-octadecane, PMMA-PETRA, MEPCM and GO/MEPCM were summarized and compared in Table 4. It was found that the measured effective thermal conductivity (k_e) of MEPCM was strongly affected by the porosity, which was the product of the low thermal conductivity of the air pores. To compare the thermal conductivity of MEPCM and GO/MEPCM samples, the real thermal conductivity (k_r) of these samples were calculated by Eq. (6) and Eq. (7), as shown in Table 4.

According to Table 4, the real thermal conductivity of n-octadecane was equal to the measured thermal conductivity of 0.34 W/m • K since the porosity of n-octadecane was zero. The real thermal conductivity of PMMA-PETRA particles was 0.10 W/m • K. According to the DSC data, the core content of MEPCM was 92 %, so the PMMA-PETRA shell weight percentage accounts for 8 %. Therefore, based on the thermal conductivity of n-octadecane (0.34 W/m • K) and PMMA-PETRA (0.10 W/m • K) and the shell volume percentages calculated by Eq. (8), the calculated theoretical thermal conductivity of MEPCM was 0.327 W/m • K which was calculated using the parallel thermal conductivity model of Kohout et al. [31]. Meanwhile, the measured real thermal conductivity of MEPCM after removing the porosity effect was 0.32 W/m • K. The measured value was therefore the same as the calculated thermal conductivity by Kohout's parallel model (with a difference of less than 0.007 W/m • K), which confirms the reliability of this model under the conditions of the study.

For the thermal conductivities of GO/MEPCM samples, the porosity was within the range of 28 % – 33 %. The measurement results showed that the thermal conductivity of GO/MEPCM was significantly increased by the increase of GO contents. After removing the influence of air porosity, the calculated thermal conductivity of GO/MEPCM was significantly increased from 0.40 W/m • K to 1.04 W/m • K when the GO content was increased from 0.1 w.t.% to 0.5 w.t.%. The respective increases of the thermal conductivity were 25 % – 225 % higher than the initial MEPCM without the added GO.

Su et al. [15] reviewed and summarized the thermal conductivity enhancement of different GO contents within MEPCM, and found that

the thermal conductivity of MEPCM reached 0.252 W/m • K and 1.067 W/m • K when 10 w.t.% GO was added within the PCM core and on the surface of microcapsules, respectively [38]. When GO was added to the surface of MEPCM particles the thermal conductivity was much higher than when it was added within the PCM core. This was because GO forms a continuous thermal pathway between the PCM core, MEPCM particles and the external environment. In this study, the thermal conductivity of GO/MEPCM-5 reached 1.04 W/m • K when 0.5 w.t.% GO was added as a co-emulsifier, which was significantly higher than the thermal conductivity of 0.28 W/m • K obtained in Liu et al.'s study when the same 0.5 w.t.% of GO was used [39]. During the emulsion polymerization of the PMMA-PETRA shell, the generated PMMA-PETRA nanobeads were blocked at the oil/water interface by GO nanosheets and then formed a GO/PMMA-PETRA hybrid shell, as shown in Fig. 1 (b). This means that GO successfully created a continuous thermal path with much less GO dosage compared with the above-mentioned previous studies.

3.7. Mechanical properties analysis

The mechanical properties of MEPCM were also important for their long-term stability. This is because of the volume shrinkage and expansion of MEPCM capsules that occur during the solidification and melting cycles. Fig. 7 shows the loading-unloading curves for the MEPCM samples. As summarized in Table 5, the indentation retention force required for the MEPCM sample was 1.30 mN but it was significantly increased when GO was added in different proportions to the GO/MEPCM capsules. However, the loading force of GO/MEPCM-3 was less than that of GO/MEPCM-1 and GO/MEPCM-2 thus indicating that GO was not uniformly dispersed on the surface of the microcapsule. As shown in SEM Fig. 4, the use of GO leads to the formation of protruding particles on the surface of the MEPCM which could result in an uneven stress distribution pattern.

In addition (Table 5), the calculated Young's modulus and hardness values with Eq. (9) and Eq. (10) for MEPCM were 0.1 GPa and 0.03 GPa, respectively. According to a previous study, Young's modulus of PDMS

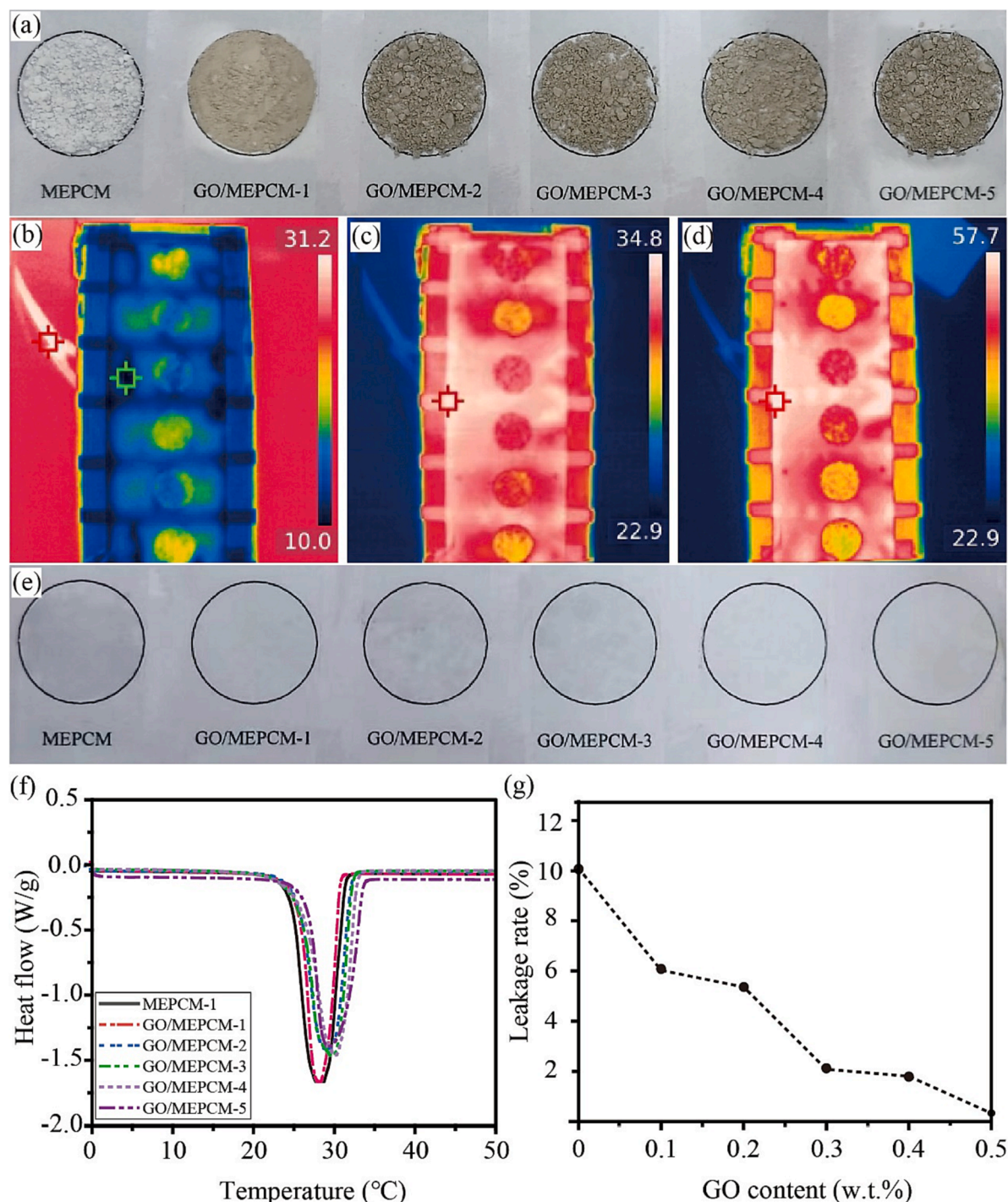


Fig. 6. MEPCMs anti-leakage experiments. (a) MEPCM samples for thermal cycles. (b) Temperature of solidified MEPCM samples. (c) Heating of MEPCM samples. (d) Temperature of melted MEPCM samples. (e) Underlay paper oil stains of MEPCMs after 100 thermal cycles. (f) DSC curves for MEPCM melting. (g) Leakage rate of MEPCM samples.

was only 1.94 MPa [40], which is much less than the testing result of our MEPCM sample. This result proved that the nanoindentation test points must be correctly loaded on the surface of the microcapsules. When the GO content was increased to 0.5 w.t.%, the Young's modulus increased to 0.3 GPa (from 0.1 GPa, 3 times of MEPCM's Young's modulus) and the hardness also increased to 0.13 GPa from 0.03 GPa, about 4.3 times of MEPCM's hardness.

Su et al. [14] developed MEPCM with a graphene/melamine-formaldehyde organic hybrid shell and then analyzed the mechanical properties of capsules by nanoindentation. The testing results

showed that Young's modulus of MEPCM samples was increased from 2.4 GPa to 4.5 GPa for the sample with 6 w.t.% graphene, which is twice the value measured without graphene. The yield strength after adding 6 w.t.% graphene was about 1.46 times (from 1.2 MPa to 1.75 MPa) that of the sample without graphene. In comparison to Su's study [14], our study only used 0.5 w.t.% GO and resulted in a three-fold increase in Young's modulus and a four-fold increase in yield strength. The mechanical properties are also directly related to the shell thickness of the microcapsule. The SEM image shows that the average particle size of the microcapsule was around 10 μm . According to the DSC results and the

Table 4
Measured and calculated thermal conductivity of materials in this study.

Sample	k_e (W/m • K)	Porosity (ϵ) (%)	k_r (W/m • K)
n-Octadecane	0.34	0	–
PMMA-PETRA	0.068	42	0.10
MEPCM	0.225	33	0.32
GO/MEPCM-1	0.285	30	0.40
GO/MEPCM-2	0.369	32	0.53
GO/MEPCM-3	0.504	28	0.69
GO/MEPCM-4	0.672	28	0.92
GO/MEPCM-5	0.706	33	1.04

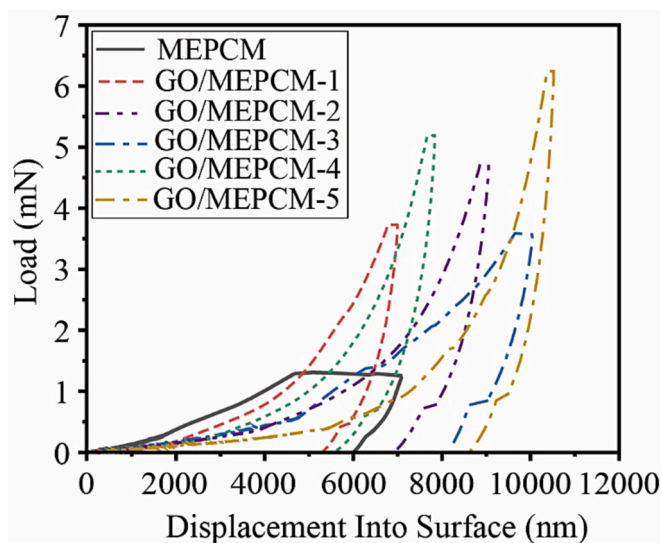


Fig. 7. Load-displacement curves of MEPCM and GO/MEPCMs.

Table 5
Micromechanical properties of MEPCM and GO/MEPCMs.

Sample	P_{max} (mN)	Theoretical shell thickness (nm)	Y (GPa)	H (GPa)
MEPCM	1.30	137	0.1	0.03
GO/MEPCM-1	3.71	245	0.2	0.02
GO/MEPCM-2	4.72	245	0.2	0.06
GO/MEPCM-3	3.58	264	0.3	0.07
GO/MEPCM-4	5.19	264	0.3	0.07
GO/MEPCM-5	6.24	397	0.3	0.13

actual shell-core ratio in Table 3, the theoretical shell thickness of the microcapsule was obtained with Eq. (3) and Eq. (4) by assuming particle size of MEPCM at 10 μ m. The calculation results in Table 5 showed that the thickness of the MEPCM shell was increased from 137 nm to 397 nm by increasing the content of GO. This may be due to the addition of GO preventing the loss of PMMA-PETRA shell monomers. Meanwhile, the accumulation of GO on the surface of the microcapsules caused secondary shell formation, further improving the shell thickness.

3.8. Comparison with previous studies and discussion

To evaluate the key parameters of MEPCM samples, a comparison with previous studies is summarized in Table 6. For instance, the latent heat and core material contents in this study were significantly higher than previous MEPCM samples fabricated with a n-octadecane core by Li

Table 6
Comparison of key parameters of the MEPCM samples.

Index	This study	Previous studies	Ref.
Latent heat (J/g)	191.1–223.3	179	[25]
		148.6–153.8	[36]
Core material content (%)	78.4–91.8	67.2–73	[25]
			[36]
Thermal stability ($^{\circ}$ C)	198–219	150–200	[30]
Leakage rate (%)	0.32 by 0.5 w.t.% GO	1.07 by 5 w.t.% GO	[41]
Thermal conductivity (W/(m•K))	1.04 by 0.5 w.t.% GO	1.067 by 10 w.t.% GO	[38]
		0.28 by 0.5 w.t.% GO	[39]
Increased Young's modulus (%)	200 % by 0.5 w.t.% GO	87.5 % by 6 w.t.% graphene	[14]
Increased yield strength (%)	333 % by 0.5 w.t.% GO	45.8 % by 6 w.t.% graphene	

et al. [25] (179.9 J/g and 67.2 %) and Zhao et al. [36] (153.82 J/g and 73 %). Compared with the MEPCM sample also fabricated with PMMA-PETRA shell by Al-Shannaq et al. [30], the thermal stability of MEPCMs in this study increased by more than 19 $^{\circ}$ C. The minimum leakage rate of MEPCMs is only 0.32 % in this study which is significantly lower than Maithya et al.'s MEPCM that was also fabricated with 5 w.t.% GO [41]. The thermal conductivity of GO/MEPCM-5 was significantly higher than the MEPCM sample obtained with 0.5 w.t.% of GO in Liu et al.'s study [39], and it was similar to the thermal conductivity of the MEPCM sample with 10 w.t.% GO by Qiao et al.'s [38]. The increased Young's modulus and yield strength of GO/MEPCMs in this study were also much higher than in a previous study by adding graphene in the MEPCM shell [14]. Overall, the resulting properties of the GO/MEPCM samples of this study were significantly improved compared to previous studies.

4. Conclusions

MEPCM samples with different contents of GO were successfully prepared by emulsion polymerization. Characterization of the samples also established the presence of GO after microencapsulation and the resulting properties were significantly improved compared to those found in previous relevant studies. The specific characterization outputs can be summarized as follows:

- 1) The GO/MEPCM samples demonstrate an improved fabrication efficiency of the capsules, particularly for the utilization rate of PMMA-PETRA shell monomers. The latent heat of GO/MEPCM with 0.5 w.t.% GO was 190.4 J/g with a 78.4 % core material content. In addition, the shell monomers utilization rate reached 72 % for the GO/MEPCM-5 sample from a baseline of less than 30 % for the MEPCM without GO.
- 2) The addition of GO within the range of 0–0.5 w.t.% reduced the supercooling of the developed MEPCM samples. The supercooling temperature of GO/MEPCM-3 was only 0.67 $^{\circ}$ C as opposed to 1.3 $^{\circ}$ C for the MEPCM sample without GO.
- 3) The TG results showed that the addition of GO further increased the weight loss temperature of the microcapsules by 7–21 $^{\circ}$ C, which shows that adding GO improves the MEPCM stability.
- 4) The addition of GO significantly improved the anti-leakage property of MEPCMs. The leakage rate was reduced from 10 % to 0.32 % by adding 0.5 w.t.% GO.
- 5) The real thermal conductivity of the GO/MEPCMs was significantly improved with the addition of GO. The thermal conductivity of the GO/MEPCM with 0.5 w.t.% GO increased by about 225 % compared with that of the MEPCM without GO.
- 6) The addition of GO also improved the mechanical properties of the microcapsule. In comparison with the initial MEPCM sample, the Young's modulus for the GO/MEPCMs increased from 0.1 GPa to 0.3 GPa whilst the hardness also increased from 0.03 GPa to 0.13 GPa for the sample of 0.5 w.t.% GO.

In conclusion, the study has shown overall improvements in the thermophysical and mechanical properties of the developed MEPCM samples containing GO. Furthermore, the obtained high thermal conductivity and mechanical strength of MEPCM samples could be considered appropriate for future practical high efficient thermal energy storage applications.

CRediT authorship contribution statement

Meiyong Hu: Investigation, Visualization, Writing – original draft, Formal analysis, Methodology. **Dawei Wang:** Methodology, Writing – original draft. **Georgios Kokogiannakis:** Writing – review & editing. **Jo Darkwa:** Funding acquisition, Writing – review & editing. **Yilin Li:** Funding acquisition, Writing – review & editing. **Li Wang:** Funding acquisition, Supervision, Writing – review & editing. **Qing Xu:** Writing – review & editing. **Weiguang Su:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Weiguang Su reports financial support was provided by Qing Chuang plan by Department of Education of Shandong Province. Li Wang reports financial support was provided by Jinan Scientific Research Leader Studio. Li Wang reports financial support was provided by Science, Education and Industry Integration Innovation Pilot Project from Qilu University of Technology. Jo Darkwa reports financial support was provided by UK Engineering, Physical Sciences Research Council (EPSRC). Yilin Li reports financial support was provided by Shanghai Sailing Program].

Data availability

The authors do not have permission to share data.

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References

- [1] Z. Chang, K. Wang, X. Wu, G. Lei, Q. Wang, H. Liu, Y. Wang, Q. Zhang, Review on the preparation and performance of paraffin-based phase change microcapsules for heat storage, *J. Storage Mater.* 46 (2022), <https://doi.org/10.1016/j.est.2021.103840>.
- [2] C.Y. Zhao, G.H. Zhang, Review on microencapsulated phase change materials (MEPCMs): Fabrication, characterization and applications, *Renew. Sustain. Energy Rev.* 15 (8) (2011) 3813–3832, <https://doi.org/10.1016/j.rser.2011.07.019>.
- [3] H. Chen, R. Zhao, C. Wang, L. Feng, S. Li, Y. Gong, Preparation and Characterization of Microencapsulated Phase Change Materials for Solar Heat Collection, *Energies* 15 (15) (2022), <https://doi.org/10.3390/en15155354>.
- [4] K. Zhao, J. Wang, H. Xie, Z. Guo, Microencapsulated phase change n-Octadecane with high heat storage for application in building energy conservation, *Appl. Energy* 329 (2023), <https://doi.org/10.1016/j.apenergy.2022.120284>.
- [5] S.N. Sonare, S.J. Jaiswal, P.A. Mahanwar, Review on applications of microencapsulated phase change material in buildings for thermal storage system, *J. Polym. Res.* 29 (9) (2022), <https://doi.org/10.1007/s10965-022-03212-1>.
- [6] C. Wang, Y. Huang, H. Wei, F. Yu, M. Wang, Y. Guo, J. Zhang, R. Deng, S. Yang, X. Liu, A brief strategy for designing self-encapsulated Al-Si base phase change materials with high thermal energy storage performance, *J. Storage Mater.* 62 (2023), <https://doi.org/10.1016/j.est.2023.106957>.
- [7] H. Liu, H. Shen, H. Zhang, X. Wang, Development of photoluminescence phase-change microcapsules for comfort thermal regulation and fluorescent recognition applications in advanced textiles, *J. Storage Mater.* 49 (2022), <https://doi.org/10.1016/j.est.2022.104158>.
- [8] W. Su, J. Darkwa, G. Kokogiannakis, Review of solid–liquid phase change materials and their encapsulation technologies, *Renew. Sustain. Energy Rev.* 48 (2015) 373–391, <https://doi.org/10.1016/j.rser.2015.04.044>.
- [9] G. Dou, M. Xu, Y. Hu, Y. Sun, H. Jiang, G. Peng, Improving the mechanical performance of phase change microcapsule/epoxy composites via alternately enhancing interface bonding and strength of microcapsules, *J. Storage Mater.* 73 (2023), 109212, <https://doi.org/10.1016/j.est.2023.109212>.
- [10] H. Hosseiniaveh, I. Rahgozar Abadi, O. Mohammadi, A. Khademi, M. Behshad Shafii, The impact of employing carbon nanotube and Fe₃O₄ nanoparticles along with intermediate boiling fluid to improve the discharge rate of phase change material, *Appl. Therm. Eng.* 215 (2022), 119032, <https://doi.org/10.1016/j.applthermaleng.2022.119032>.
- [11] T. Xiong, K.W. Shah, H.W. Kua, Application of graphite platelets for heat transfer enhancement of cementitious composites containing microencapsulated phase change materials, *Constr. Build. Mater.* 318 (2022), <https://doi.org/10.1016/j.conbuildmat.2021.126024>.
- [12] A. Khezri, M. Sahebi, M. Mohammadi, Fabrication and Thermal properties of graphene nanoplatelet-enhanced phase change materials based on paraffin encapsulated by melamine–formaldehyde, *J. Therm. Anal. Calorim.* (2021), <https://doi.org/10.1007/s10973-021-11085-7>.
- [13] Y. Chen, G. Yang, L. Chen, B. Wang, X. Feng, Z. Mao, X. Sui, H. Xu, Phase change microcapsules with graphene nanoplates embedded polyurea shell for enhanced thermal conductivity, *Mater. Lett.* 330 (2023), <https://doi.org/10.1016/j.matlet.2022.133223>.
- [14] J.-F. Su, X.-Y. Wang, S. Han, X.-L. Zhang, Y.-D. Guo, Y.-Y. Wang, Y.-Q. Tan, N.-X. Han, W. Li, Preparation and physicochemical properties of microcapsules containing phase-change material with graphene/organic hybrid structure shells, *J. Mater. Chem. A* 5 (45) (2017) 23937–23951, <https://doi.org/10.1039/C7TA06980D>.
- [15] W. Su, M. Hu, L. Wang, G. Kokogiannakis, J. Chen, L. Gao, A. Li, C. Xu, Microencapsulated phase change materials with graphene-based materials: Fabrication, characterisation and prospects, *Renew. Sustain. Energy Rev.* 168 (2022), <https://doi.org/10.1016/j.rser.2022.112806>.
- [16] L. Zhang, K. Wu, Y. Chen, R. Liu, J. Luo, The preparation of linseed oil loaded graphene/polyaniline microcapsule via emulsion template method for self-healing anticorrosion coatings, *Colloid. Surf. A: Physicochem. Eng. Aspect.* 651 (2022), <https://doi.org/10.1016/j.colsurfa.2022.129771>.
- [17] L. Zhang, W. Yang, Z. Jiang, F. He, K. Zhang, J. Fan, J. Wu, Graphene oxide-modified microencapsulated phase change materials with high encapsulation capacity and enhanced leakage-prevention performance, *Appl. Energy* 197 (2017) 354–363, <https://doi.org/10.1016/j.apenergy.2017.04.041>.
- [18] Z. Jiang, W. Yang, F. He, C. Xie, J. Fan, J. Wu, K. Zhang, Modified Phase Change Microcapsules with Calcium Carbonate and Graphene Oxide Shells for Enhanced Energy Storage and Leakage Prevention, *ACS Sustain. Chem. Eng.* (2018), <https://doi.org/10.1021/acssuschemeng.7b04834>.
- [19] G. Dou, M. Xu, Y. Hu, Y. Sun, H. Jiang, G. Peng, Improving the mechanical performance of phase change microcapsule/epoxy composites via alternately enhancing interface bonding and strength of microcapsules, *J. Storage Mater.* 73 (2023), <https://doi.org/10.1016/j.est.2023.109212>.
- [20] J. Xu, S. Chen, J. Wang, J. Du, X. Dong, T. Zhang, X. Ren, G. Liang, F. Zhang, X. Li, H. Yin, C. Liu, Y. Ke, Y. Zhao, Y. Ding, Octadecane-Encapsulated, Polyurethane PolyHIPE Composites with Flexibility and Stretchability for Personal Heat Management, *ACS Appl. Polym. Mater.* 5 (10) (2023) 7927–7935, <https://doi.org/10.1021/acsaapm.3c01167>.
- [21] L. Zhu, W. Ding, F. Dang, G. Sang, Y. Xue, Q. Wang, Combined experimental and theoretical investigation of pore characteristics effect on thermal conductivity of light-weight aggregate concrete including microencapsulated PCM, *J. Mater. Res. Technol.* 26 (2023) 587–602, <https://doi.org/10.1016/j.jmrt.2023.07.226>.
- [22] L. Xu, K. Zhang, R. He, A. Yang, L. Su, Y. Li, F. He, S. Jiang, W. Yang, Phase change composites based on double-shell microcapsules with high latent heat and low leakage rate for thermal energy storage and temperature regulation, *J. Storage Mater.* 55 (2022), <https://doi.org/10.1016/j.est.2022.105428>.
- [23] K. Xu, Y. Luo, Packaging ink microcapsules with high stability and biocompatibility based on natural dye gardenia blue, *Colloids Surf A Physicochem Eng Asp* 658 (2023), <https://doi.org/10.1016/j.colsurfa.2022.130778>.
- [24] H.Y. Woo, D.W. Lee, T.Y. Yoon, J.B. Kim, J.Y. Chae, T. Paik, Sub-100-nm Nearly Monodisperse n-Paraffin/PMMA Phase Change Nanobeads, *Nanomaterials* 11 (1) (2021), <https://doi.org/10.3390/nano11010204>.
- [25] C. Li, H. Yu, Y. Song, H. Liang, X. Yan, Preparation and characterization of PMMA/TiO₂ hybrid shell microencapsulated PCMs for thermal energy storage, *Energy* 167 (2019) 1031–1039, <https://doi.org/10.1016/j.energy.2018.11.038>.
- [26] X. Qiu, W. Li, G. Song, X. Chu, G. Tang, Fabrication and characterization of microencapsulated n-octadecane with different crosslinked methylmethacrylate-based polymer shells, *Sol. Energy Mater. Sol. Cells* 98 (2012) 283–293, <https://doi.org/10.1016/j.solmat.2011.11.018>.
- [27] C. Alkan, A. Sari, A. Karaipekli, O. Uzun, Preparation, characterization, and thermal properties of microencapsulated phase change material for thermal energy storage, *Sol. Energy Mater. Sol. Cells* 93 (1) (2009) 143–147, <https://doi.org/10.1016/j.solmat.2008.09.009>.

- [28] X. Qiu, L. Lu, Z. Chen, Preparation and characterization of flame retardant phase change materials by microencapsulated paraffin and diethyl ethylphosphonate with poly(methacrylic acid-co-ethyl methacrylate) shell, *J. Appl. Polym. Sci.* 132 (17) (2015), <https://doi.org/10.1002/app.41880>.
- [29] H. Wang, L. Zhao, G. Song, G. Tang, X. Shi, Organic-inorganic hybrid shell microencapsulated phase change materials prepared from SiO₂/TiC-stabilized pickering emulsion polymerization, *Sol. Energy Mater. Sol. Cells* 175 (2018) 102–110, <https://doi.org/10.1016/j.solmat.2017.09.015>.
- [30] R. Al-Shannaq, M. Farid, S. Al-Muhtaseb, J. Kurdi, Emulsion stability and cross-linking of PMMA microcapsules containing phase change materials, *Sol. Energy Mater. Sol. Cells* 132 (2015) 311–318, <https://doi.org/10.1016/j.solmat.2014.08.036>.
- [31] J. Darkwa, O. Su, T. Zhou, Development of non-deform micro-encapsulated phase change energy storage tablets, *Appl. Energy* 98 (2012) 441–447, <https://doi.org/10.1016/j.apenergy.2012.04.006>.
- [32] Y.-T. Huang, H. Zhang, X.-J. Wan, D.-Z. Chen, X.-F. Chen, X. Ye, X. Ouyang, S.-Y. Qin, H.-X. Wen, J.-N. Tang, Carbon nanotube-enhanced double-walled phase-change microcapsules for thermal energy storage, *J. Mater. Chem. A* 5 (16) (2017) 7482–7493, <https://doi.org/10.1039/c6ta09712j>.
- [33] Y.-D. Guo, J.-F. Su, R. Mu, X.-Y. Wang, X.-L. Zhang, X.-M. Xie, Y.-Y. Wang, Y.-Q. Tan, Microstructure and Properties of Self-Assembly Graphene Microcapsules: Effect of the pH Value, *Nanomaterials* 9 (4) (2019) 587, <https://doi.org/10.3390/nano9040587>.
- [34] A. Malas, C.K. Das, Effect of graphene oxide on the physical, mechanical and thermo-mechanical properties of neoprene and chlorosulfonated polyethylene vulcanizates, *Compos. B Eng.* 79 (2015) 639–648, <https://doi.org/10.1016/j.compositesb.2015.04.051>.
- [35] M. Zhao, M. Li, L. Wang, X. Zhang, X. Kong, Preparation and Characterization of Paraffin@CLPS/MS Phase Change Microcapsules for Thermal Energy Storage, *ChemistrySelect* 5 (24) (2020) 7190–7196, <https://doi.org/10.1002/slct.202001263>.
- [36] J. Zhao, Y. Yang, Y. Li, L. Zhao, H. Wang, G. Song, G. Tang, Microencapsulated phase change materials with TiO₂-doped PMMA shell for thermal energy storage and UV-shielding, *Sol. Energy Mater. Sol. Cells* 168 (2017) 62–68, <https://doi.org/10.1016/j.solmat.2017.04.014>.
- [37] B. Xie, G. Liu, S. Jiang, Y. Zhao, D. Wang, Crystallization Behaviors of n-Octadecane in Confined Space: Crossover of Rotator Phase from Transient to Metastable Induced by Surface Freezing, *J. Phys. Chem. B* 112 (42) (2008) 13310–13315, <https://doi.org/10.1021/jp712160k>.
- [38] Z. Qiao, J. Mao, Enhanced thermal properties with graphene oxide in the urea-formaldehyde microcapsules containing paraffin PCMs, *J. Microencapsul.* 34 (1) (2017) 1–9, <https://doi.org/10.1080/02652048.2016.1267811>.
- [39] Z. Liu, Z. Chen, F. Yu, Enhanced thermal conductivity of microencapsulated phase change materials based on graphene oxide and carbon nanotube hybrid filler, *Sol. Energy Mater. Sol. Cells* 192 (2019) 72–80, <https://doi.org/10.1016/j.solmat.2018.12.014>.
- [40] J.M. Kim, F. Wolf, S.K. Baier, Effect of varying mixing ratio of PDMS on the consistency of the soft-contact Stribeck curve for glycerol solutions, *Tribol. Int.* 89 (2015) 46–53, <https://doi.org/10.1016/j.triboint.2014.12.010>.
- [41] O.M. Maithya, X. Li, X. Feng, X. Sui, B. Wang, Microencapsulated phase change material via Pickering emulsion stabilized by graphene oxide for photothermal conversion, *J. Mater. Sci.* (2020), <https://doi.org/10.1007/s10853-020-04499-5>.